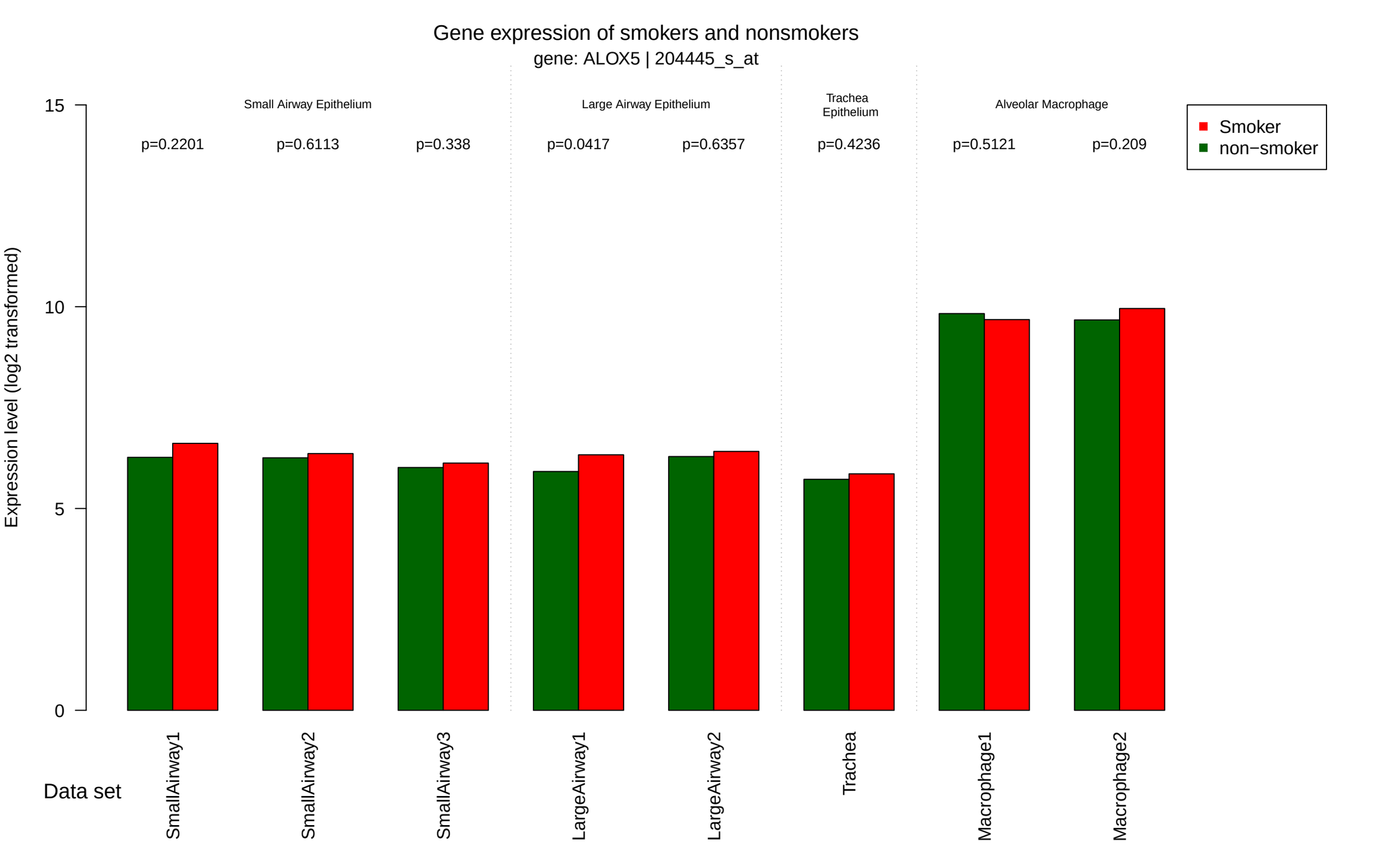
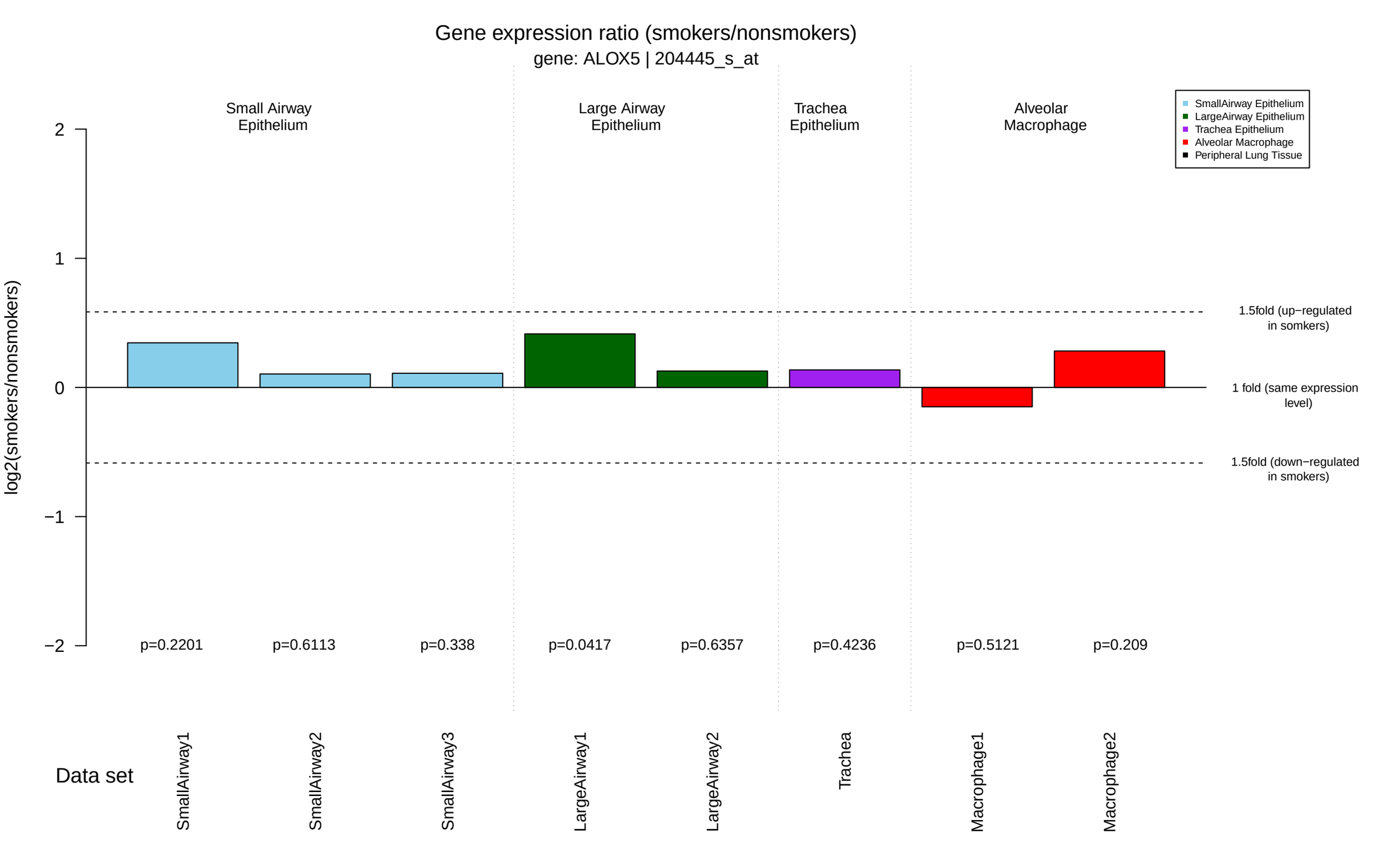
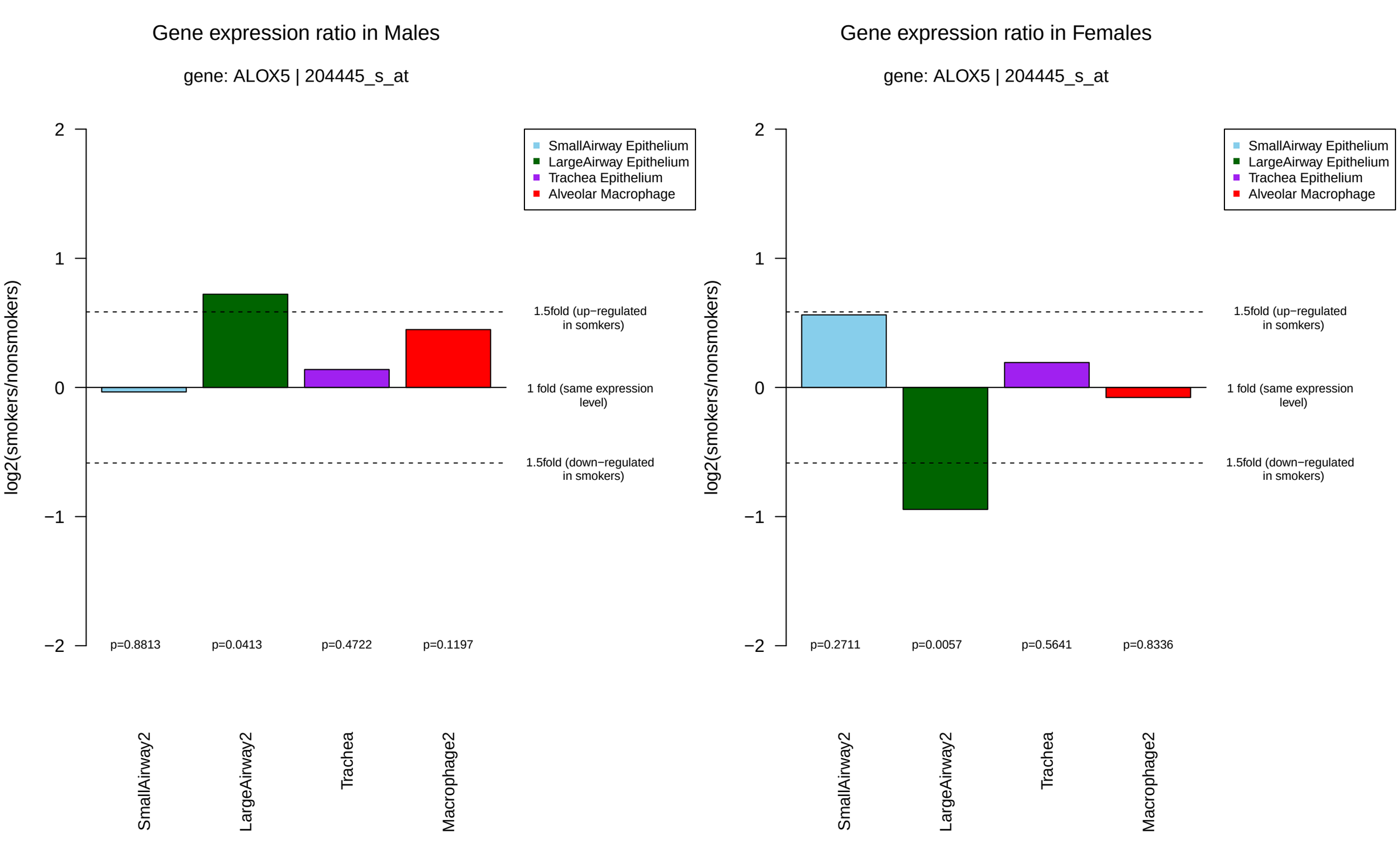
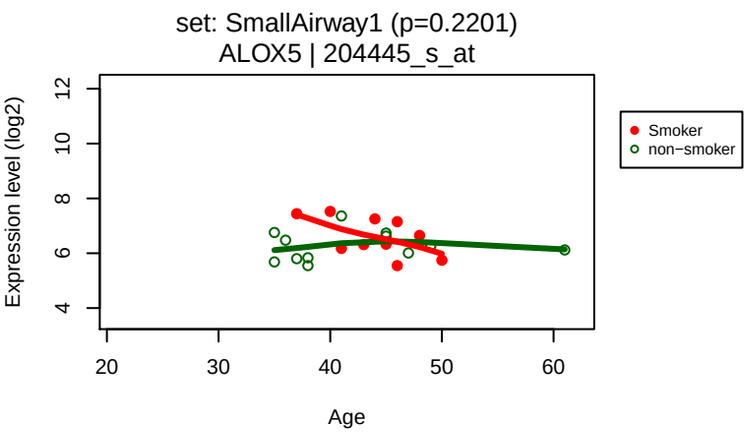
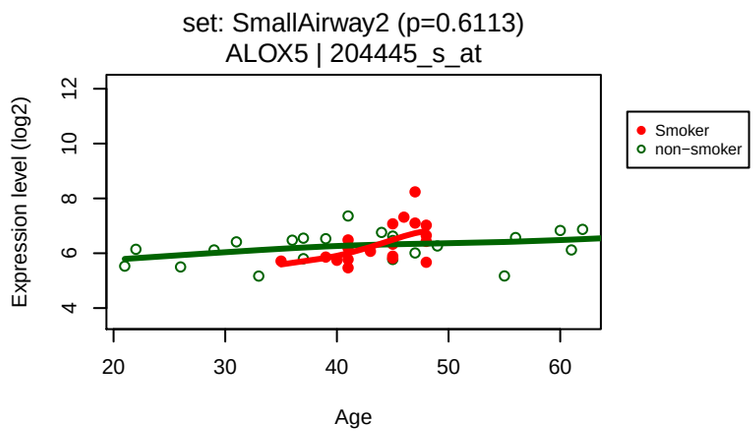
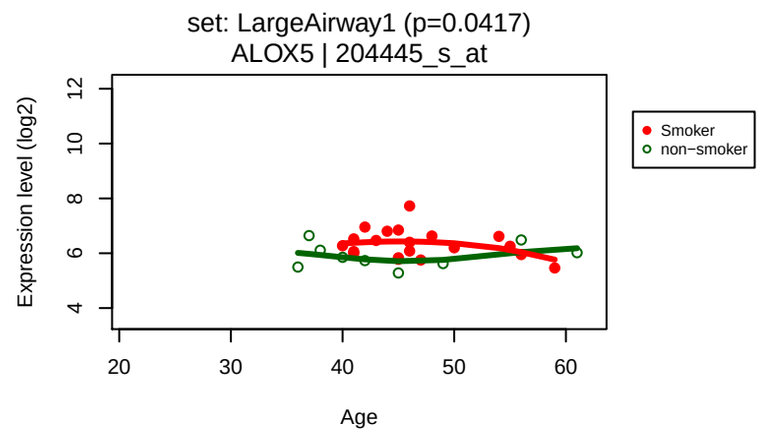

Gene expression of smokers and nonsmokers
gene: ALOX5 | 204445_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2201
p=0.6113
p=0.338
p=0.0417
p=0.6357
p=0.4236
p=0.5121
p=0.209
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

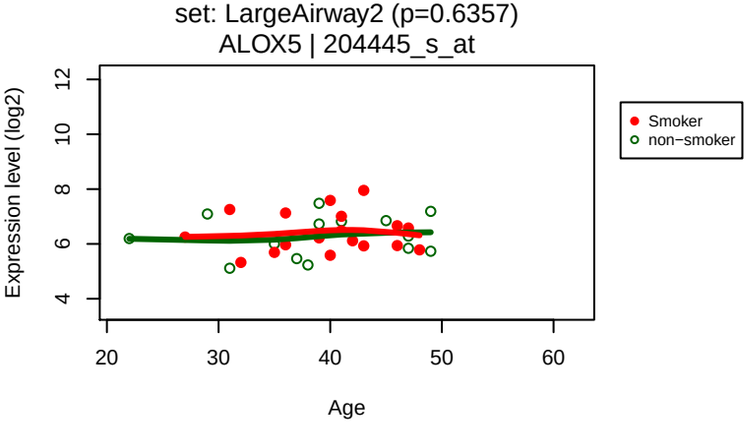

Gene expression ratio (smokers/nonsmokers)
gene: ALOX5 | 204445_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2201
p=0.6113
p=0.338
p=0.0417
p=0.6357
p=0.4236
p=0.5121
p=0.209
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

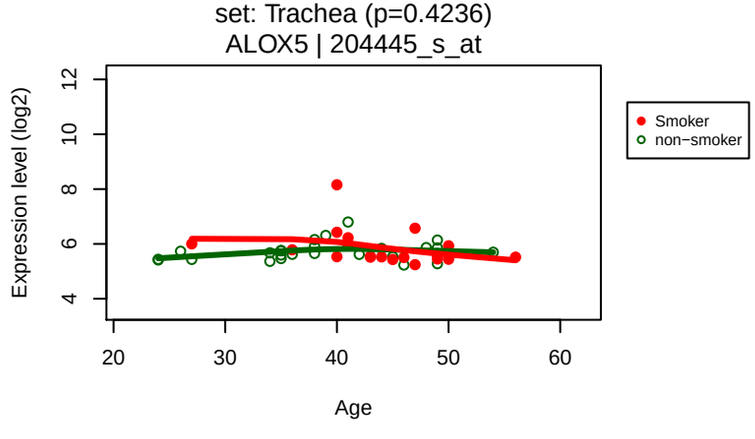

Gene expression ratio in Males
gene: ALOX5 | 204445_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8813
p=0.0413
p=0.4722
p=0.1197
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOX5 | 204445_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2711
p=0.0057
p=0.5641
p=0.8336
SmallAirway2
LargeAirway2
Trachea
Macrophage2

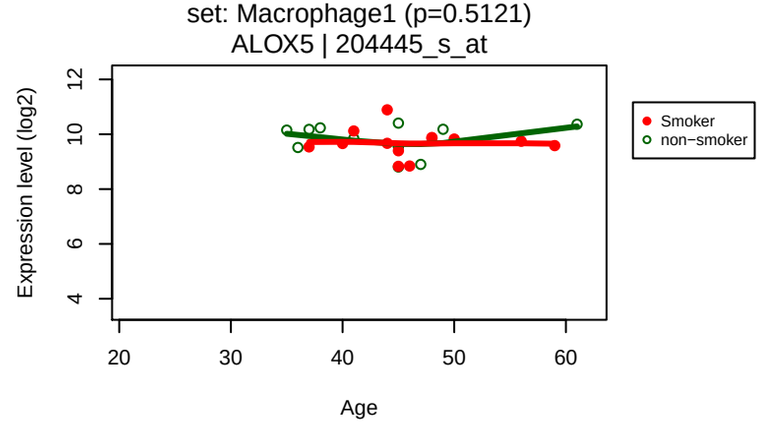

set: SmallAirway1 (p=0.2201)
ALOX5 | 204445_s_at
Expression level (log2)
Age
Smoker
non−smoker

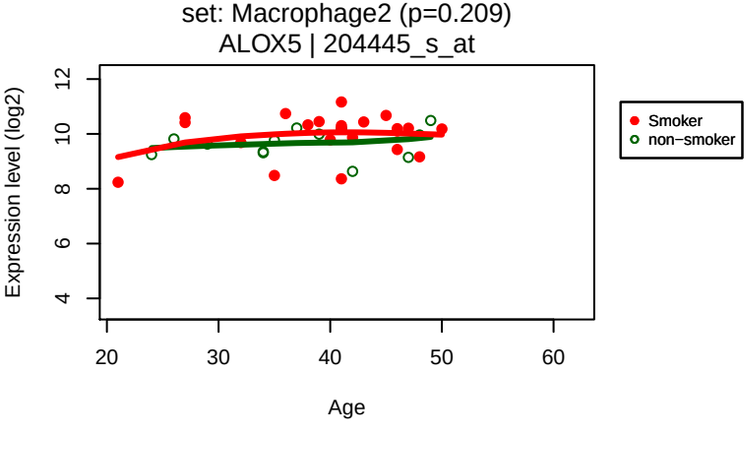

set: SmallAirway2 (p=0.6113)
ALOX5 | 204445_s_at
Expression level (log2)
Age
Smoker
non−smoker

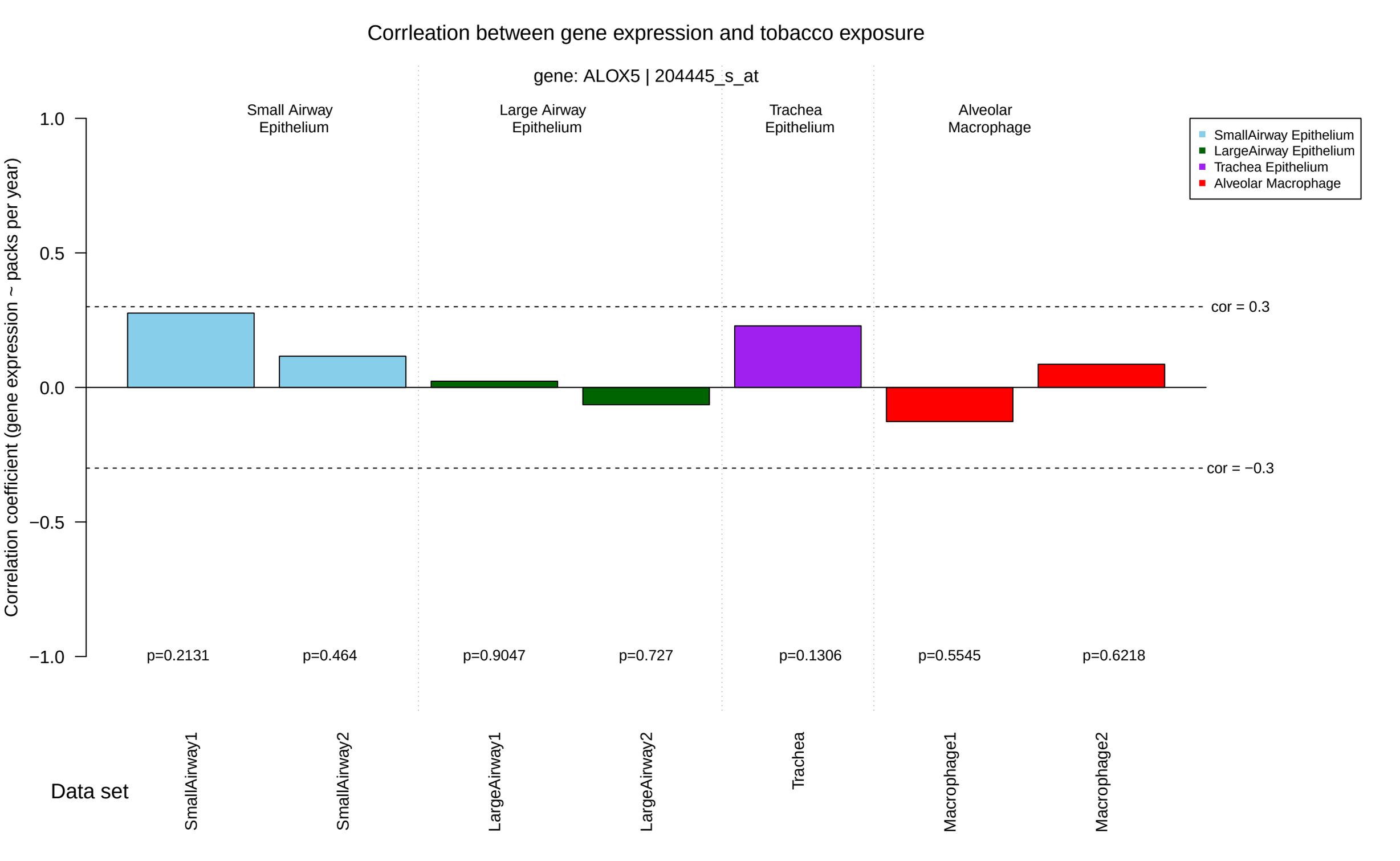

set: LargeAirway1 (p=0.0417)
ALOX5 | 204445_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6357)
ALOX5 | 204445_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4236)
ALOX5 | 204445_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5121)
ALOX5 | 204445_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.209)
ALOX5 | 204445_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOX5 | 204445_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2131
p=0.464
p=0.9047
p=0.727
p=0.1306
p=0.5545
p=0.6218
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2